Second Partial

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Derivative of logarithms.

Today's session we are going to learn how to compute the derivative of logartihms functions.
The overall mthodology is as follows:

1. Use the change base formula
2. Compute the derivative of the change base formula
3. If possible, return to original base

Example: $f(x) = \log(x)$

Use the change base formula

$$\log(x) = \frac{\ln(x)}{\ln(10)}$$

Example: $f(x) = \log(x)$

Compute the derivative

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\frac{\ln(x)}{\ln(10)} &= \frac{1}{\ln(10)}\frac{\mathrm{d}}{\mathrm{d}x}\ln(x) \\
&= \frac{1}{\ln(10)}\frac{1}{x} \\
&= \frac{1}{\ln(10)x}
\end{aligned}$$

Example: $f(x) = \log(x)$

Return to original base

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\frac{\ln(x)}{\ln(10)} &= \frac{1}{\ln(10)x} \\
&= \frac{1}{\frac{1}{\log(e)}x} \\
&= \frac{\log(e)}{x}
\end{aligned}
$$

Example: $f(x) = \log(x)$

Answers

$$\boxed{f'(x) = \frac{1}{\ln(10)x} = \frac{\log(e)}{x}}$$

Class Activity

Find the derivatives of the following logaritmic functions,

$$f(x) = \log_{\pi}(x)$$
$$g(x) = \log_{2}(2^{x} + 1)$$